MSc. in Computing
Practicum Approval Form

Project Title:	Automated Assessment of Consent Dialogues for GDPR Compliance
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Date of Submission:	28/11/2024

Topic:

The practicum discusses the challenge of automating GDPR compliance assessment in the cookie consent dialogues using the advanced data analytics and natural language processing techniques. Our research focused on development of an intelligent system which can extract, analyze, and standardize information from various consent mechanisms across the websites. By combining traditional web scraping with cutting-edge language models (including the transformer-based NLP and LLMs), we focus to create a comprehensive analysis framework for the compliance assessment. The system will transform complex legal terminology and different dialogue structures into standardized, analyzable formats at the same time preserving legal meaning. Through the data-driven insights and automated analysis, this practicum seeks to enhance transparency in data collection practices and provides the practical tools for both the end users and the organizations to better understand and evaluate privacy compliance in real-time.

Research Questions:

How can we design an automated system to analyze cookie consent dialogues and assess the GDPR compliance through the data-driven approaches and natural language processing? These consent dialogues often use complex legal terminology, and they are having an inconsistent structure across websites, which makes it challenging for both users and organizations to understand privacy implications. This will be focused on how we can use the NLP techniques to transform these unstructured dialogues into standardized, analyzable formats while preserving the legal meaning, ultimately helping users making informed decisions about their own privacy choices and assisting organizations while maintaining compliance.

Primary Research Objectives:

1. **Data Collection and Analytics: -**

Developing an automated system for extraction and analysis of cookie consent information across different websites, by taking help of web scraping and data analytics to analyze the different consent patterns. This approach is inspired by techniques for compliance assessment from IAB Europe's TCF.

2. **Standardization and Processing: -**

Transforming diverse consent dialogues into a well-structured format using standardized privacy vocabularies, mapping with frameworks like DPV-2. This provides the consistent representation of consent information for automated analysis of extracted data.

3. **Machine Learning Implementation: -**

To build and train the transformer-based NLP models which can comprehend complex legal terminology in the consent dialogues, building on previous work in privacy policy analysis using similar architectures.

4. **Real-World Application Development: -**

To create practical tools that make privacy compliance information accessible, including a browser extension for real-time analysis. This leverages insights from compliance metrics studies to provide user-friendly interfaces for both users and privacy professionals.

We have analyzed several key studies that are relevant to our research. These studies explore:

- Pandit, H. J., Polleres, A., De, S., Bos, B., Lizar, M., Gerl, A., ... & Wenning, R. (2023). Data Privacy Vocabulary (DPV): Version 2. Semantic Web Journal. https://arxiv.org/pdf/2404.13426
  Provides standardized vocabulary for metadata representation and compliance checking

- Nouwens, M., Liccardi, I., Veale, M., Karger, D., & Kagal, L. (2020). Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, 1-14.
  https://arxiv.org/pdf/2001.02479
  Demonstrates techniques for consent dialogue scraping and dark pattern identification

- Matte, C., Bielova, N., & Santos, C. (2020). Do Cookie Banners Respect my Choice? Measuring Legal Compliance of Banners from IAB Europe's Transparency and Consent Framework. In Proceedings of The Web Conference 2020 (WWW '20) (pp. 2091-2102). ACM.
  https://arxiv.org/pdf/1911.09964

Offers methodology for compliance assessment and metrics

- Gray, C. M., Santos, C., Bielova, N., Toth, M., & Clifford, D. (2021). Dark Patterns and the Legal Requirements of Consent Banners: An Interaction Criticism Perspective. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, Article 172, 1–18. https://dl.acm.org/doi/pdf/10.1145/3411764.3445779
  Provides framework for analyzing dark patterns in consent mechanisms

- Zhang, L., Moukafih, N., Alamri, H., Epiphaniou, G., & Maple, C. (2024). A BERT-based empirical study of privacy policies' compliance with GDPR. arXiv:2407.06778. https://arxiv.org/pdf/2407.06778
  Shows effectiveness of transformer models for privacy compliance analysis

Relation to Existing Work:

Our project proposal extends existing research by combining the automated metadata extraction with NLP-based compliance analysis system. The Data Privacy Vocabulary (DPV) – version 2.0 paper provides the vocabulary for training our NLP model. The dark patterns research informs our model's feature engineering for detecting problematic practices in data collection. The IAB Europe's Transparency and Consent Framework compliance study offers metrics for evaluating consent mechanisms. The BERT-based privacy policy study demonstrates the effectiveness of transformer models for privacy text analysis.

Our project innovates by:

- Developing an NLP model specifically for consent dialogue analysis
- Creating automated compliance checking using transformer architecture
- Implementing real-time analysis through browser extension
- Providing standardized metadata representation using DPV
- Contributing to privacy-focused NLP research

Exploration Methodology

Technology and Implementation:

- We will use Python, leveraging its robust libraries for the research goals:
- Web scraping using Selenium and BeautifulSoup to extract the consent dialogues dynamically from various websites
- Hugging Face Transformers library and PyTorch in the development and training of our NLP model, inspired by successful privacy policy analysis work.
- Pandas and scikit-learn to structure, analyze, and process consent information into standardized formats.

Data Collection and Processing:

- Scrapped Consent Dialogues: We will automate the gathering of data from various websites by consent assessment methodologies.
- Annotated Compliance Datasets: These will feed our model to train and be accurate in detecting compliance.

- Privacy-related policy documents: To understand the legal framework for compliance analysis.
- DPV-2 Vocabulary: dpv-2-vocabulary.html: for standardized metadata representation.

Data Analytics Focus:

Our project will analyze consent patterns, compliance trends, and identify common issues across websites, contributing to the data analytics in privacy research.

Experiments and Validation:

- We will train and validate our NLP model on annotated consent dialogues.
- Compliance analysis accuracy will be checked by cross-validation from human experts.
- We'll do pattern recognition testing, on the very subtle issues of compliance.

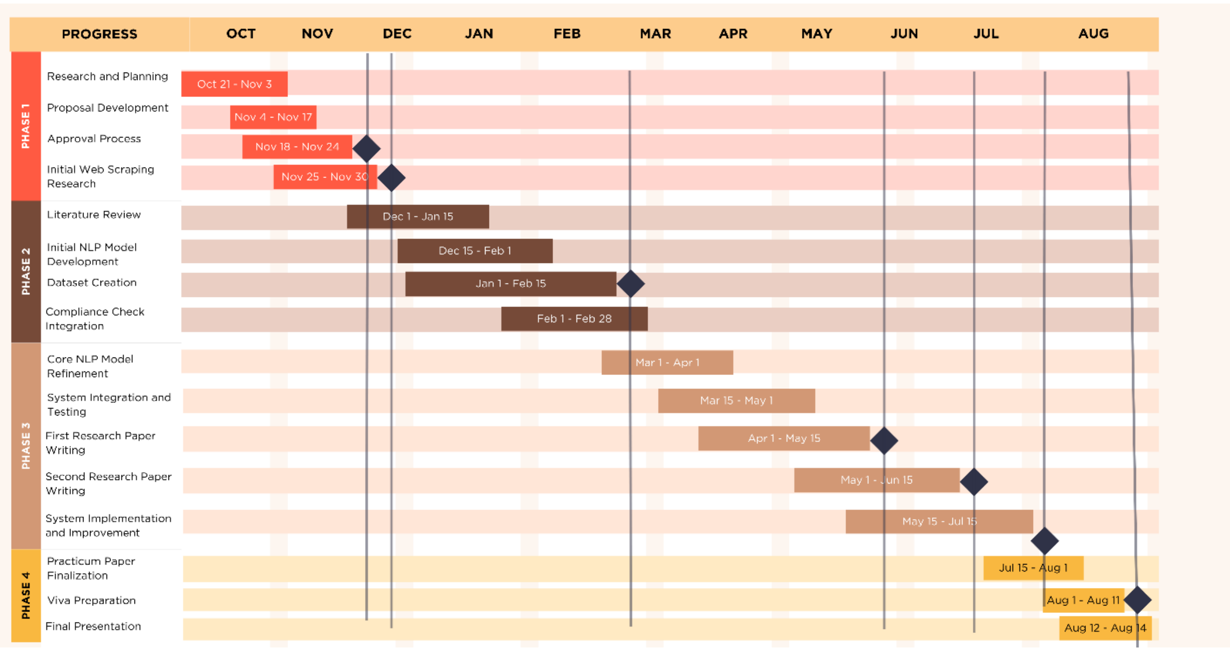
Expected Output:

- Trained NLP Model for consent dialogue analysis.
- Real-time intelligence: Compliance Analysis System.
- Structured Metadata for further analysis.
- Browser Extension for practical user and organizational use.

Evaluation:

- We'll measure our model's performance using NLP metrics.
- Compliance accuracy will be validated against legal standards.
- Expert review will ensure our findings align with privacy regulations.
- User testing will validate the utility of our browser extension.

Gantt Chart



Milestones (◆) :

1. Submit approval form: End of November 2024 (Phase 1)
2. Present practicum proposal: End of November 2024 (Phase 1)
3. Submit literature review: Mid-February 2025 (Phase 2)
4. Submit first research paper: Mid-May 2025 (Phase 3)
5. Submit second research paper: Mid-June 2025 (Phase 3)
6. Submit practicum papers: Late July 2025 (Between Phase 3 and 4)
7. Conduct practicum viva: Mid-August 2025 (Phase 4)

Timeline

November 2024:

Develop initial web scraping scripts.

Monthly Milestone: Submission of practicum proposal.

Present practicum proposal to the panel.

December 2024:

Begin testing NLP model with sample data.

Monthly Milestone: Focus on exams until 13th December.

January 2025:

Resume project work post-holidays.

Continue developing and testing NLP model.

Monthly Milestone: Draft literature review sections.

February 2025:

Finalize first draft of literature review.

Implement compliance checks in the model.

Monthly Milestone: Submit literature review.

March 2025:

Implement model improvements based on evaluation.

Begin outlining research papers.

Monthly Milestone: Finalize core NLP model functionality.

April 2025:

Perform user testing (if applicable).

Monthly Milestone: Complete first research paper draft.

Address system integration issues.

May 2025:

Monthly Milestone: First research paper submission.

Start writing the second research paper.

Preparation of initial viva presentation draft.

June 2025:

Monthly Milestone: The second research paper submission to the peer-reviewing journal.

Editing and finalization of the practicum papers.

July 2025:

Monthly Milestone: Paper presentation in conferences.

Conduct mock viva sessions.

August 2025:

Monthly Milestone: Viva Practicum to be conducted on 12th - 14th August.

Plan for post-viva research paper revisions or submissions.

Note: - Drafted this timeline based on the Practicum schedule and classes/exams schedules.